DOMAIN GUIDE

Loan Terms And Commitments

Every part of the product must refer to the same loan. A borrower and a selected lender can see an offer, authorize keys and recipients in their Vaults, move USDC on Base, and prepare collateral transactions on Bitcoin; those artifacts must agree about the participants, amounts, deadlines, destinations, and secret commitments. This domain owns that shared meaning. A matching identifier proves which agreement an artifact describes; it does not authorize a spend or establish a chain outcome.

> This is the domain guide for Loan Terms And Commitments, one of the five domains of Bitcoin-backed USDC one-loan system. The architecture guide, a separate document, explains how the domains work together.

A **firm offer** records the lender's proposed commercial values. A **reservation** selects one offer revision for one borrower and creates a stable random `loan_id`. **LoanTermsV1** is the ordered, versioned agreement data; its `terms_hash` binds the exact encoded values. **ProtocolDeploymentV1** is different: it identifies the approved environment, chain, contracts, asset, Bitcoin network, and policy configuration. URLs, credentials, secrets, and mutable application state are not part of that deployment identity. Separating immutable loan facts from environment facts prevents different consumers from signing or processing different meanings under the same label.

One Responsibility, Shared Meaning

Loan terms, one-loan offer, reservation, and parameter binding is the domain's single responsibility. It defines identity, encoding, candidate versions, approvals, commitment, cancellation, and supersession. Its output is used by every other domain: Bitcoin construction binds the package graph to it; Base settlement accepts only the pinned terms and deployment; observers reject wrong-deployment evidence; participant operations persist and present the same candidate; Vaults check the same role and loan binding before returning public identities or signing. The loan terms and canonical binding contract, the protocol deployment manifest, and the offer, reservation, and loan binding contract own the exact representations and public boundaries.

The detailed records distinguish commercial reservation from final agreement. Reservation creates the loan identity before loan-specific Vault derivation. Both roles then supply durable acknowledged public allocation results. The borrower prepares its public destination and `H(X)` result; Core assembles a complete candidate version and hash. Each role explicitly approves that same candidate, and the atomic second matching approval commits it once. A hashed-field edit invalidates earlier candidate approvals. Duplicate identical approvals and commitment retries return the same result; stale, wrong-role, wrong-loan, mismatched, expired, or cancelled approvals are rejected. The offer and reservation states, the canonical loan terms states, and the origination flow preserve that ordering. The architecture record instead states that publication is

the lender's standing acceptance of the offer and that reservation creates the canonical loan without a second lender approval. This is an unresolved source conflict: neither wording is chosen until it is answered, so whether the lender approves the final candidate separately remains open.

Boundaries And Exceptions

A protocol Bitcoin public key, Base protocol externally owned account (EOA), external payout destination, and Vault payment account are distinct identities. The payment account can fund an eligible transfer; it does not silently become a loan recipient or protocol caller. Selecting an address for a fresh loan's existing recipient field uses the normal role authorization and fresh candidate approvals. Historical terms remain immutable. Likewise, an acceptance snapshot or workspace copy cannot become a second authority for agreement meaning. The canonical loan binding, selected lender capital control, authority separation, and key, destination, and refund separation invariants make those boundaries explicit.

Every consumer rejects an unsupported encoding, changed fingerprint, wrong chain/contract/asset/network, mixed environment, wrong key or destination, or dependent artifact that does not bind exactly. Pre-activation amendment produces a later candidate/version and fresh dependent artifacts. Cancellation ends that reservation; reopening requires a fresh loan identity. These rules prevent an apparently successful retry from reviving old approval or signing material.

Choices And Scope

The recorded rationale for canonical encoding is interoperability and verifiability: divergent encodings or mixed immutable and deployment facts could bind signatures, manifests, and chain actions to different loans, or make a package impossible to reproduce. The exact encoding and accepted address spellings remain in the loan terms and canonical binding contract; the domain explanation does not duplicate that byte-level contract.

The lender separately retains the capital-funding decision, even after an agreement is committed. Product authentication, Vault decryption, Bitcoin construction, Base execution, observation, and durable workflow coordination remain with their respective domains. Public or multi-lender matching, pooled capital, product custody, and automatic deployment of lender capital remain excluded under the matching, pooling, custody, and automatic capital exclusion, and liquidation under the liquidation, surplus, and shortfall exclusion. An unresolved implementation question belongs in its existing ticket; a narrative or diagram cannot supply a missing approval.

The architecture guide shows where this domain fits in the system, and the domain record lists exactly what the domain owns and excludes.

Other guides

This guide refers to these separate documents of the System Guide:

- How the Bitcoin-backed USDC one-loan system works (architecture guide)

Technical records

The exact rules, contracts and conditions live in these records of the design package:

- **Domain record:** Loan Terms And Commitments.
- **Responsibility record:** Loan Terms, One-Loan Offer, Reservation, And Parameter Binding.
- **Flow record:** Origination Preparation And Qualification.
- **Interface contracts:** LoanTermsV1 And Canonical Binding; One-Loan Offer, Reservation, And Loan Binding; ProtocolDeploymentV1.
- **State contracts:** Canonical Loan Terms; One-Loan Offer And Reservation.
- **Invariants:** Authority Separation; Canonical Loan Binding; Key, Destination, And Refund Separation; Selected Lender Capital Control And No Liquidation.
- **Scope boundaries:** BTC Liquidation, Automated Surplus Return, And Shortfall Enforcement; Public Or Multi-Lender Matching, Pooling, Custody, And Automatic Capital Deployment.